

# Appendix

## Full Simulation Tables for the Primary Functionals

### **Appendix. Full simulation tables for the primary functionals**

The main text summarizes the principal simulation findings and shows coverage across conditions. This appendix provides the complete numerical results for the three primary functionals— $ab$ ,  $\psi_{\text{speed}}$ , and  $\omega_{\text{speed}}$ —at  $N = 100, 200$ , and  $500$ , including coverage, left- and right-tail noncoverage, and median interval width.

**Table 1:** Simulation results for the indirect effect  $ab$  at  $N = 100$ : coverage of nominal 95% intervals, tail balance, and median width, by condition and method ( $S = 1,000$  data sets per cell,  $B = 1,000$  shared bootstrap weight vectors).

| Method                                     | Coverage | Miss left | Miss right | Median width |
|--------------------------------------------|----------|-----------|------------|--------------|
| <i>Normal data, correct model</i>          |          |           |            |              |
| Wald (Expected)                            | 89.7     | 0.2       | 10.1       | 0.216        |
| Wald (HW)                                  | 90.0     | 0.1       | 9.9        | 0.229        |
| Monte Carlo (HW)                           | 94.0     | 0.3       | 5.7        | 0.246        |
| IJ1 percentile                             | 94.0     | 0.3       | 5.7        | 0.242        |
| HOIJ-2 percentile                          | 94.9     | 0.6       | 4.5        | 0.251        |
| Bootstrap percentile                       | 95.7     | 0.5       | 3.8        | 0.255        |
| <i>Normal data, misspecified model</i>     |          |           |            |              |
| Wald (Expected)                            | 90.0     | 0.4       | 9.6        | 0.226        |
| Wald (HW)                                  | 91.2     | 0.6       | 8.2        | 0.239        |
| Monte Carlo (HW)                           | 93.7     | 0.9       | 5.4        | 0.257        |
| IJ1 percentile                             | 94.0     | 0.9       | 5.1        | 0.256        |
| HOIJ-2 percentile                          | 94.5     | 1.4       | 4.1        | 0.266        |
| Bootstrap percentile                       | 95.4     | 1.4       | 3.3        | 0.269        |
| <i>Non-normal data, correct model</i>      |          |           |            |              |
| Wald (Expected)                            | 88.8     | 0.3       | 10.9       | 0.210        |
| Wald (HW)                                  | 90.2     | 0.3       | 9.5        | 0.219        |
| Monte Carlo (HW)                           | 93.5     | 0.5       | 6.0        | 0.237        |
| IJ1 percentile                             | 92.9     | 0.6       | 6.5        | 0.235        |
| HOIJ-2 percentile                          | 94.3     | 0.9       | 4.8        | 0.239        |
| Bootstrap percentile                       | 96.0     | 0.9       | 3.0        | 0.246        |
| <i>Non-normal data, misspecified model</i> |          |           |            |              |
| Wald (Expected)                            | 89.3     | 0.7       | 10.0       | 0.219        |
| Wald (HW)                                  | 90.3     | 0.7       | 9.0        | 0.230        |
| Monte Carlo (HW)                           | 93.0     | 0.8       | 6.2        | 0.245        |
| IJ1 percentile                             | 93.0     | 0.7       | 6.3        | 0.245        |
| HOIJ-2 percentile                          | 94.1     | 1.4       | 4.5        | 0.250        |
| Bootstrap percentile                       | 95.2     | 1.2       | 3.6        | 0.253        |

*Note.* Coverage, miss-left, and miss-right entries are percentages, based on  $S = 1,000$  simulated data sets per cell. For a well-calibrated equal-tailed 95% interval, the reference noncoverage rates are 2.5% in each tail. Differences between HOIJ-2 and the bootstrap describe differences between the procedures; differences from 2.5% describe inferential tail imbalance.

**Table 2:** Simulation results for the indirect effect  $ab$  at  $N = 200$ .

| Method                                     | Coverage | Miss left | Miss right | Median width |
|--------------------------------------------|----------|-----------|------------|--------------|
| <i>Normal data, correct model</i>          |          |           |            |              |
| Wald (Expected)                            | 92.0     | 0.4       | 7.6        | 0.153        |
| Wald (HW)                                  | 92.8     | 0.4       | 6.8        | 0.156        |
| Monte Carlo (HW)                           | 95.0     | 1.0       | 4.0        | 0.161        |
| IJ1 percentile                             | 95.0     | 0.6       | 4.4        | 0.159        |
| HOIJ-2 percentile                          | 94.7     | 1.2       | 4.1        | 0.161        |
| Bootstrap percentile                       | 95.8     | 1.1       | 3.1        | 0.165        |
| <i>Normal data, misspecified model</i>     |          |           |            |              |
| Wald (Expected)                            | 92.9     | 0.5       | 6.6        | 0.158        |
| Wald (HW)                                  | 94.0     | 0.6       | 5.4        | 0.162        |
| Monte Carlo (HW)                           | 95.4     | 0.9       | 3.7        | 0.166        |
| IJ1 percentile                             | 95.7     | 0.9       | 3.4        | 0.166        |
| HOIJ-2 percentile                          | 95.3     | 1.6       | 3.1        | 0.167        |
| Bootstrap percentile                       | 95.7     | 1.4       | 2.9        | 0.170        |
| <i>Non-normal data, correct model</i>      |          |           |            |              |
| Wald (Expected)                            | 91.3     | 0.9       | 7.8        | 0.154        |
| Wald (HW)                                  | 91.7     | 0.3       | 8.0        | 0.159        |
| Monte Carlo (HW)                           | 93.4     | 0.7       | 5.9        | 0.163        |
| IJ1 percentile                             | 92.7     | 1.1       | 6.2        | 0.163        |
| HOIJ-2 percentile                          | 93.2     | 1.5       | 5.3        | 0.163        |
| Bootstrap percentile                       | 94.0     | 1.4       | 4.6        | 0.167        |
| <i>Non-normal data, misspecified model</i> |          |           |            |              |
| Wald (Expected)                            | 90.8     | 0.2       | 9.0        | 0.157        |
| Wald (HW)                                  | 91.6     | 0.1       | 8.3        | 0.165        |
| Monte Carlo (HW)                           | 93.7     | 0.6       | 5.7        | 0.168        |
| IJ1 percentile                             | 93.8     | 0.4       | 5.8        | 0.168        |
| HOIJ-2 percentile                          | 94.9     | 0.8       | 4.3        | 0.167        |
| Bootstrap percentile                       | 95.8     | 0.6       | 3.6        | 0.170        |

Note. See Table 1.

**Table 3:** Simulation results for the indirect effect  $ab$  at  $N = 500$ .

| Method                                     | Coverage | Miss left | Miss right | Median width |
|--------------------------------------------|----------|-----------|------------|--------------|
| <i>Normal data, correct model</i>          |          |           |            |              |
| Wald (Expected)                            | 93.1     | 0.9       | 6.0        | 0.096        |
| Wald (HW)                                  | 93.3     | 0.9       | 5.8        | 0.098        |
| Monte Carlo (HW)                           | 94.2     | 1.6       | 4.2        | 0.098        |
| IJ1 percentile                             | 93.7     | 1.6       | 4.7        | 0.097        |
| HOIJ-2 percentile                          | 94.4     | 1.6       | 4.0        | 0.098        |
| Bootstrap percentile                       | 94.7     | 1.7       | 3.6        | 0.099        |
| <i>Normal data, misspecified model</i>     |          |           |            |              |
| Wald (Expected)                            | 94.2     | 1.0       | 4.8        | 0.099        |
| Wald (HW)                                  | 94.5     | 1.0       | 4.5        | 0.099        |
| Monte Carlo (HW)                           | 95.5     | 1.1       | 3.4        | 0.099        |
| IJ1 percentile                             | 95.0     | 1.2       | 3.8        | 0.099        |
| HOIJ-2 percentile                          | 95.6     | 1.3       | 3.1        | 0.099        |
| Bootstrap percentile                       | 96.1     | 1.2       | 2.7        | 0.100        |
| <i>Non-normal data, correct model</i>      |          |           |            |              |
| Wald (Expected)                            | 92.2     | 1.5       | 6.3        | 0.095        |
| Wald (HW)                                  | 93.0     | 0.8       | 6.2        | 0.099        |
| Monte Carlo (HW)                           | 94.1     | 1.2       | 4.7        | 0.099        |
| IJ1 percentile                             | 93.4     | 1.6       | 5.0        | 0.099        |
| HOIJ-2 percentile                          | 93.7     | 2.0       | 4.3        | 0.099        |
| Bootstrap percentile                       | 94.0     | 2.0       | 4.0        | 0.100        |
| <i>Non-normal data, misspecified model</i> |          |           |            |              |
| Wald (Expected)                            | 92.2     | 1.6       | 6.2        | 0.099        |
| Wald (HW)                                  | 93.5     | 0.9       | 5.6        | 0.102        |
| Monte Carlo (HW)                           | 94.7     | 1.3       | 4.0        | 0.103        |
| IJ1 percentile                             | 94.1     | 1.4       | 4.5        | 0.103        |
| HOIJ-2 percentile                          | 94.2     | 2.1       | 3.7        | 0.102        |
| Bootstrap percentile                       | 94.7     | 2.0       | 3.3        | 0.104        |

Note. See Table 1.

**Table 4:** Simulation results for the structural disturbance variance  $\psi_{\text{speed}}$  at  $N = 100$ : coverage of nominal 95% intervals, tail balance, and median width, by condition and method ( $S = 1,000$  data sets per cell,  $B = 1,000$  shared bootstrap weight vectors).

| Method                                     | Coverage | Miss left | Miss right | Median width |
|--------------------------------------------|----------|-----------|------------|--------------|
| <i>Normal data, correct model</i>          |          |           |            |              |
| Wald (Expected)                            | 91.8     | 0.4       | 7.8        | 0.516        |
| Wald (HW)                                  | 92.2     | 0.6       | 7.2        | 0.525        |
| Monte Carlo (HW)                           | 91.9     | 0.7       | 7.4        | 0.526        |
| IJ1 percentile                             | 92.2     | 0.5       | 7.3        | 0.527        |
| HOIJ-2 percentile                          | 94.9     | 1.1       | 4.0        | 0.549        |
| Bootstrap percentile                       | 94.9     | 1.1       | 4.1        | 0.531        |
| <i>Normal data, misspecified model</i>     |          |           |            |              |
| Wald (Expected)                            | 92.8     | 0.4       | 6.8        | 0.510        |
| Wald (HW)                                  | 92.3     | 1.0       | 6.7        | 0.521        |
| Monte Carlo (HW)                           | 92.3     | 1.1       | 6.6        | 0.514        |
| IJ1 percentile                             | 92.3     | 0.9       | 6.8        | 0.517        |
| HOIJ-2 percentile                          | 94.4     | 1.3       | 4.3        | 0.539        |
| Bootstrap percentile                       | 95.0     | 1.4       | 3.7        | 0.528        |
| <i>Non-normal data, correct model</i>      |          |           |            |              |
| Wald (Expected)                            | 88.7     | 1.4       | 9.9        | 0.496        |
| Wald (HW)                                  | 89.2     | 0.2       | 10.6       | 0.513        |
| Monte Carlo (HW)                           | 89.0     | 0.3       | 10.7       | 0.509        |
| IJ1 percentile                             | 88.7     | 0.6       | 10.7       | 0.509        |
| HOIJ-2 percentile                          | 92.0     | 1.2       | 6.8        | 0.534        |
| Bootstrap percentile                       | 93.5     | 1.0       | 5.5        | 0.518        |
| <i>Non-normal data, misspecified model</i> |          |           |            |              |
| Wald (Expected)                            | 90.1     | 1.8       | 8.1        | 0.487        |
| Wald (HW)                                  | 90.2     | 0.3       | 9.5        | 0.493        |
| Monte Carlo (HW)                           | 90.1     | 0.3       | 9.6        | 0.491        |
| IJ1 percentile                             | 89.5     | 0.6       | 9.9        | 0.490        |
| HOIJ-2 percentile                          | 92.7     | 1.2       | 6.1        | 0.518        |
| Bootstrap percentile                       | 93.6     | 0.8       | 5.7        | 0.493        |

Note. See Table 1.

**Table 5:** Simulation results for the structural disturbance variance  $\psi_{\text{speed}}$  at  $N = 200$ .

| Method                                     | Coverage | Miss left | Miss right | Median width |
|--------------------------------------------|----------|-----------|------------|--------------|
| <i>Normal data, correct model</i>          |          |           |            |              |
| Wald (Expected)                            | 92.6     | 1.0       | 6.4        | 0.372        |
| Wald (HW)                                  | 93.3     | 0.8       | 5.9        | 0.373        |
| Monte Carlo (HW)                           | 93.0     | 1.0       | 6.0        | 0.371        |
| IJ1 percentile                             | 93.2     | 1.0       | 5.8        | 0.371        |
| HOIJ-2 percentile                          | 95.2     | 1.7       | 3.1        | 0.377        |
| Bootstrap percentile                       | 95.4     | 1.4       | 3.2        | 0.374        |
| <i>Normal data, misspecified model</i>     |          |           |            |              |
| Wald (Expected)                            | 92.5     | 0.8       | 6.7        | 0.371        |
| Wald (HW)                                  | 92.3     | 0.8       | 6.9        | 0.375        |
| Monte Carlo (HW)                           | 92.0     | 1.1       | 6.9        | 0.373        |
| IJ1 percentile                             | 92.0     | 1.2       | 6.8        | 0.374        |
| HOIJ-2 percentile                          | 93.7     | 1.9       | 4.4        | 0.377        |
| Bootstrap percentile                       | 93.7     | 1.7       | 4.6        | 0.374        |
| <i>Non-normal data, correct model</i>      |          |           |            |              |
| Wald (Expected)                            | 90.7     | 0.8       | 8.5        | 0.368        |
| Wald (HW)                                  | 89.7     | 0.2       | 10.1       | 0.371        |
| Monte Carlo (HW)                           | 90.6     | 0.2       | 9.2        | 0.372        |
| IJ1 percentile                             | 89.3     | 0.6       | 10.1       | 0.372        |
| HOIJ-2 percentile                          | 92.8     | 1.0       | 6.2        | 0.381        |
| Bootstrap percentile                       | 92.7     | 0.9       | 6.4        | 0.378        |
| <i>Non-normal data, misspecified model</i> |          |           |            |              |
| Wald (Expected)                            | 92.1     | 1.4       | 6.5        | 0.361        |
| Wald (HW)                                  | 91.6     | 0.3       | 8.1        | 0.366        |
| Monte Carlo (HW)                           | 91.5     | 0.3       | 8.2        | 0.364        |
| IJ1 percentile                             | 91.2     | 0.6       | 8.2        | 0.366        |
| HOIJ-2 percentile                          | 93.8     | 1.3       | 4.9        | 0.372        |
| Bootstrap percentile                       | 94.3     | 1.1       | 4.6        | 0.369        |

Note. See Table 1.

**Table 6:** Simulation results for the structural disturbance variance  $\psi_{\text{speed}}$  at  $N = 500$ .

| Method                                     | Coverage | Miss left | Miss right | Median width |
|--------------------------------------------|----------|-----------|------------|--------------|
| <i>Normal data, correct model</i>          |          |           |            |              |
| Wald (Expected)                            | 95.2     | 1.1       | 3.7        | 0.241        |
| Wald (HW)                                  | 95.3     | 1.0       | 3.7        | 0.243        |
| Monte Carlo (HW)                           | 95.1     | 1.2       | 3.7        | 0.241        |
| IJ1 percentile                             | 94.5     | 1.2       | 4.3        | 0.242        |
| HOIJ-2 percentile                          | 95.1     | 1.9       | 3.0        | 0.243        |
| Bootstrap percentile                       | 94.9     | 1.8       | 3.3        | 0.242        |
| <i>Normal data, misspecified model</i>     |          |           |            |              |
| Wald (Expected)                            | 93.7     | 2.1       | 4.2        | 0.243        |
| Wald (HW)                                  | 93.5     | 2.1       | 4.4        | 0.241        |
| Monte Carlo (HW)                           | 93.2     | 2.2       | 4.6        | 0.238        |
| IJ1 percentile                             | 93.1     | 2.2       | 4.7        | 0.238        |
| HOIJ-2 percentile                          | 94.2     | 2.8       | 3.0        | 0.239        |
| Bootstrap percentile                       | 94.3     | 2.7       | 3.0        | 0.238        |
| <i>Non-normal data, correct model</i>      |          |           |            |              |
| Wald (Expected)                            | 92.7     | 1.6       | 5.7        | 0.240        |
| Wald (HW)                                  | 92.9     | 0.8       | 6.3        | 0.248        |
| Monte Carlo (HW)                           | 92.1     | 1.0       | 6.9        | 0.246        |
| IJ1 percentile                             | 92.7     | 1.1       | 6.2        | 0.244        |
| HOIJ-2 percentile                          | 93.6     | 1.4       | 5.0        | 0.247        |
| Bootstrap percentile                       | 93.9     | 1.4       | 4.7        | 0.247        |
| <i>Non-normal data, misspecified model</i> |          |           |            |              |
| Wald (Expected)                            | 93.6     | 1.8       | 4.6        | 0.238        |
| Wald (HW)                                  | 94.6     | 0.8       | 4.6        | 0.241        |
| Monte Carlo (HW)                           | 94.5     | 0.9       | 4.6        | 0.242        |
| IJ1 percentile                             | 94.3     | 1.2       | 4.5        | 0.240        |
| HOIJ-2 percentile                          | 94.8     | 1.5       | 3.7        | 0.242        |
| Bootstrap percentile                       | 94.7     | 1.5       | 3.8        | 0.242        |

Note. See Table 1.

**Table 7:** Simulation results for the reliability coefficient  $\omega_{\text{speed}}$  at  $N = 100$ : coverage of nominal 95% intervals, tail balance, and median width, by condition and method ( $S = 1,000$  data sets per cell,  $B = 1,000$  shared bootstrap weight vectors).

| Method                                     | Coverage | Miss left | Miss right | Median width |
|--------------------------------------------|----------|-----------|------------|--------------|
| <i>Normal data, correct model</i>          |          |           |            |              |
| Wald (Expected)                            | 94.3     | 4.8       | 0.9        | 0.205        |
| Wald (HW)                                  | 94.3     | 4.7       | 1.0        | 0.203        |
| Monte Carlo (HW)                           | 97.4     | 0.8       | 1.8        | 0.312        |
| IJ1 percentile                             | 97.0     | 0.9       | 2.1        | 0.312        |
| HOIJ-2 percentile                          | 98.3     | 1.6       | 0.1        | 0.263        |
| Bootstrap percentile                       | 95.2     | 3.3       | 1.6        | 0.205        |
| <i>Normal data, misspecified model</i>     |          |           |            |              |
| Wald (Expected)                            | 92.9     | 5.7       | 1.4        | 0.205        |
| Wald (HW)                                  | 92.8     | 6.2       | 1.0        | 0.202        |
| Monte Carlo (HW)                           | 97.6     | 0.8       | 1.6        | 0.302        |
| IJ1 percentile                             | 97.7     | 1.1       | 1.2        | 0.296        |
| HOIJ-2 percentile                          | 98.0     | 1.9       | 0.1        | 0.256        |
| Bootstrap percentile                       | 93.9     | 4.0       | 2.1        | 0.202        |
| <i>Non-normal data, correct model</i>      |          |           |            |              |
| Wald (Expected)                            | 88.5     | 10.1      | 1.4        | 0.202        |
| Wald (HW)                                  | 91.2     | 8.2       | 0.6        | 0.217        |
| Monte Carlo (HW)                           | 97.3     | 2.4       | 0.3        | 0.321        |
| IJ1 percentile                             | 96.8     | 2.4       | 0.8        | 0.305        |
| HOIJ-2 percentile                          | 94.9     | 4.9       | 0.2        | 0.265        |
| Bootstrap percentile                       | 92.9     | 6.7       | 0.5        | 0.217        |
| <i>Non-normal data, misspecified model</i> |          |           |            |              |
| Wald (Expected)                            | 86.2     | 11.8      | 2.0        | 0.199        |
| Wald (HW)                                  | 89.3     | 10.2      | 0.5        | 0.216        |
| Monte Carlo (HW)                           | 97.0     | 2.5       | 0.5        | 0.319        |
| IJ1 percentile                             | 96.6     | 2.7       | 0.7        | 0.298        |
| HOIJ-2 percentile                          | 94.2     | 5.8       | 0.0        | 0.262        |
| Bootstrap percentile                       | 90.6     | 8.8       | 0.7        | 0.212        |

Note. See Table 1.



**Table 8:** Simulation results for the reliability coefficient  $\omega_{\text{speed}}$  at  $N = 200$ .

| Method                                     | Coverage | Miss left | Miss right | Median width |
|--------------------------------------------|----------|-----------|------------|--------------|
| <i>Normal data, correct model</i>          |          |           |            |              |
| Wald (Expected)                            | 94.2     | 4.3       | 1.5        | 0.148        |
| Wald (HW)                                  | 93.8     | 4.6       | 1.6        | 0.146        |
| Monte Carlo (HW)                           | 94.3     | 1.1       | 4.6        | 0.177        |
| IJ1 percentile                             | 93.9     | 1.3       | 4.8        | 0.175        |
| HOIJ-2 percentile                          | 96.1     | 2.7       | 1.2        | 0.160        |
| Bootstrap percentile                       | 94.1     | 3.3       | 2.6        | 0.146        |
| <i>Normal data, misspecified model</i>     |          |           |            |              |
| Wald (Expected)                            | 96.7     | 2.4       | 0.9        | 0.148        |
| Wald (HW)                                  | 96.2     | 2.5       | 1.3        | 0.147        |
| Monte Carlo (HW)                           | 94.5     | 1.0       | 4.5        | 0.177        |
| IJ1 percentile                             | 94.8     | 0.9       | 4.3        | 0.176        |
| HOIJ-2 percentile                          | 97.0     | 1.8       | 1.2        | 0.162        |
| Bootstrap percentile                       | 95.8     | 2.1       | 2.1        | 0.148        |
| <i>Non-normal data, correct model</i>      |          |           |            |              |
| Wald (Expected)                            | 89.4     | 7.0       | 3.6        | 0.147        |
| Wald (HW)                                  | 93.2     | 5.2       | 1.6        | 0.162        |
| Monte Carlo (HW)                           | 96.1     | 1.9       | 2.0        | 0.193        |
| IJ1 percentile                             | 95.3     | 2.1       | 2.6        | 0.187        |
| HOIJ-2 percentile                          | 94.6     | 4.5       | 0.9        | 0.173        |
| Bootstrap percentile                       | 93.1     | 4.9       | 1.9        | 0.162        |
| <i>Non-normal data, misspecified model</i> |          |           |            |              |
| Wald (Expected)                            | 88.9     | 8.4       | 2.7        | 0.145        |
| Wald (HW)                                  | 92.6     | 6.8       | 0.6        | 0.162        |
| Monte Carlo (HW)                           | 96.3     | 2.8       | 0.9        | 0.193        |
| IJ1 percentile                             | 95.6     | 2.8       | 1.6        | 0.187        |
| HOIJ-2 percentile                          | 94.5     | 5.3       | 0.2        | 0.173        |
| Bootstrap percentile                       | 93.4     | 5.8       | 0.8        | 0.161        |

Note. See Table 1.

**Table 9:** Simulation results for the reliability coefficient  $\omega_{\text{speed}}$  at  $N = 500$ .

| Method                                     | Coverage | Miss left | Miss right | Median width |
|--------------------------------------------|----------|-----------|------------|--------------|
| <i>Normal data, correct model</i>          |          |           |            |              |
| Wald (Expected)                            | 93.6     | 4.5       | 1.9        | 0.093        |
| Wald (HW)                                  | 93.9     | 4.3       | 1.8        | 0.093        |
| Monte Carlo (HW)                           | 94.0     | 1.8       | 4.2        | 0.100        |
| IJ1 percentile                             | 93.8     | 2.0       | 4.2        | 0.099        |
| HOIJ-2 percentile                          | 95.0     | 3.0       | 2.0        | 0.096        |
| Bootstrap percentile                       | 93.8     | 3.5       | 2.7        | 0.093        |
| <i>Normal data, misspecified model</i>     |          |           |            |              |
| Wald (Expected)                            | 95.3     | 3.0       | 1.7        | 0.093        |
| Wald (HW)                                  | 95.2     | 3.2       | 1.6        | 0.093        |
| Monte Carlo (HW)                           | 95.7     | 1.2       | 3.1        | 0.099        |
| IJ1 percentile                             | 95.5     | 1.5       | 3.0        | 0.099        |
| HOIJ-2 percentile                          | 95.9     | 2.4       | 1.7        | 0.095        |
| Bootstrap percentile                       | 95.2     | 2.8       | 2.0        | 0.093        |
| <i>Non-normal data, correct model</i>      |          |           |            |              |
| Wald (Expected)                            | 89.8     | 6.4       | 3.8        | 0.093        |
| Wald (HW)                                  | 94.4     | 4.0       | 1.6        | 0.107        |
| Monte Carlo (HW)                           | 95.4     | 2.5       | 2.1        | 0.114        |
| IJ1 percentile                             | 94.8     | 2.4       | 2.8        | 0.113        |
| HOIJ-2 percentile                          | 94.8     | 3.9       | 1.3        | 0.108        |
| Bootstrap percentile                       | 94.7     | 4.0       | 1.3        | 0.106        |
| <i>Non-normal data, misspecified model</i> |          |           |            |              |
| Wald (Expected)                            | 89.1     | 7.5       | 3.4        | 0.093        |
| Wald (HW)                                  | 94.3     | 4.7       | 1.0        | 0.106        |
| Monte Carlo (HW)                           | 95.7     | 2.9       | 1.4        | 0.113        |
| IJ1 percentile                             | 95.5     | 2.5       | 2.0        | 0.111        |
| HOIJ-2 percentile                          | 94.8     | 3.9       | 1.3        | 0.107        |
| Bootstrap percentile                       | 93.8     | 4.6       | 1.6        | 0.105        |

Note. See Table 1.